Respiratory Rate Radar Detection with Continuous Wave with Pluto SDR at 5.95GHz

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***Abstract*—This paper details the design and implementation of a contactless respiratory-rate monitor utilizing a continuous-wave (CW) Doppler radar system built on the ADALM-Pluto software-defined radio (SDR). Operating at 5.95 GHz, the system leverages the phase modulation of sub-millimeter chest displacements to estimate vital signs at standoff distances of 0.5–2m. The technical architecture features a robust signal processing chain implemented in GNU Radio, which includes a 3200-tap DC blocker to mitigate transmit-receive leakage and a phase-demodulation pipeline. To ensure high reliability, the system employs a consensus layer that fuses outputs from three concurrent estimators: FFT peak detection, zero-crossing, and autocorrelation. Additionally, a Schmitt-trigger motion detector suppresses readings during bulk subject movement to maintain data integrity.**

***Index Terms*—ADALM-Pluto, Software Defined Radio(SDR), breaths per Minute (BrPM) GNU Radio, PyQt5**

I. Introduction

Contactless vital-signs [1] sensing has become an active research area in radar engineering because conventional physiological monitoring depends on wearable transducers that subjects routinely remove, dislodge, or cannot operate [2]–[4]. Infants, post-operative sleepers, and cognitively impaired patients are the canonical examples, but the problem generalizes: any time a contact sensor interferes with the activity it is meant to monitor, a non-contact alternative has clinical value. Radio-frequency (RF) sensing meets this need with three structural advantages over contact instruments:

Material Penetration: Electromagnetic waves at microwave frequencies penetrate clothing and bedding without attenuation that matters for the physiological signal of interest.

Privacy Preservation: Radar is privacy-preserving in a way video cannot be, as no image of the subject is captured and the raw signal is a one-dimensional time series that cannot be reconstructed into a recognizable scene.

Phase Sensitivity: The relevant physiological motions—thoracic excursion during respiration and cardiac wall motion—are easily resolved by the round-trip phase of a reflected microwave carrier. The relationship between displacement Δx and phase $\Delta\phi$ is defined as:

$$\Delta\phi = \frac{4\pi \cdot \Delta x}{\lambda}$$

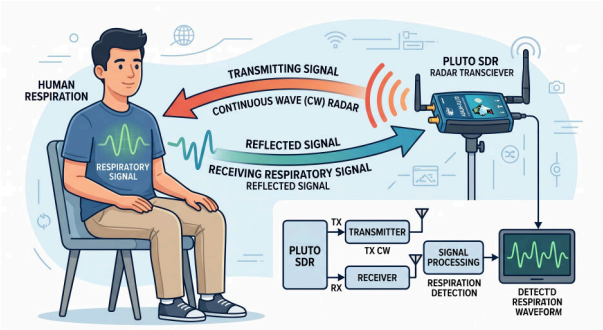


Fig. 1. Operational concept of the CW respiratory radar. The PlutoSDR transmits a 5.95 GHz tone; the reflected signal carries sub-millimeter chest-wall motion as a phase modulation that is recovered by digital signal processing to produce a respiration waveform.

where λ is the free-space wavelength of the carrier.

This paper describes a complete end-to-end respiratory-rate sensor built as a term project for a graduate radar systems course. The system is implemented on commercial off-the-shelf hardware - a single ADALM-Pluto SDR—and its entire signal chain executes in GNU Radio [5] on a commodity host computer.

The contributions of this work include documenting SDR platform suitability for coherent CW links at 5.95 GHz, modifying the conventional arctangent-unwrap pipeline for high transmit-receive leakage, and the implementation of a consensus layer to fuse independent BrPM estimators. The BrPM estimation logic draws upon established radar processing heritage, including pulse-Doppler peak detection [6] , FM discrimination [7] , and Pulse Pair Processing [6] adapted for slower physiological processes.

II. System Overview

The system overview encompasses a full hardware-to-software pipeline designed for non-invasive vital sign monitoring. At the front end, the system utilizes a Software Defined Radio (SDR) architecture, specifically the ADALM-PLUTO, to establish a high-frequency sensing environment. The process begins with the generation of a continuous wave (CW) tone at 5.95 GHz. This signal is radiated via a directional Vivaldi antenna (see fig. 2 and fig. 3) toward the subject's chest

cavity. As the subject breathes, the physical displacement of the chest wall acts as a mechanical modulator, reflecting the signal with a phase shift that is directly proportional to the minute changes in the distance between the radar and the body (see fig. 1) .Upon receiving the reflected echoes, the system performs a coherent down-conversion to translate the high-frequency signals into baseband In-phase (I) and Quadrature (Q) components.

These signals are streamed to a host computer where they undergo a multi-stage digital signal processing (DSP) routine. This involves DC-offset compensation to remove "clutter" from stationary objects in the room, followed by phase demodulation. To ensure the integrity of the data, the system extracts the phase information which captures the rhythmic oscillation of the respiratory cycle, effectively turning the chest wall's physical motion into a digital time-series waveform.

To ensure the reliability of the extracted vitals, the system implements a multi-modal consensus method that aggregates data from three parallel estimation engines: Fast Fourier Transform (FFT) for spectral peak detection, Autocorrelation (AC) for periodicity analysis, and Zero-Crossing (ZC) for low-latency time-domain tracking.

III. Hardware Platform Evaluation

Three SDR platforms were evaluated as candidate CW transceivers for this system. The selection criteria, prioritized in descending order, were as follows: (i) sufficient receiver ADC resolution to ensure that cardiac phase modulation remains above the quantization noise floor, (ii) phase coherence between transmit and receive chains without external clock hardware, (iii) reliable data streaming to the host computer at the target sampling rate and (iv) a tuning range extending to up to 6 GHz ISM band.

A. HackRF One (rejected)

The HackRF One [8] is an open-source half-duplex SDR covering 1 MHz to 6 GHz. A key challenge in using two HackRF devices as a coherent transmit–receive link is that each operates with an independent local oscillator. As a result, the relative phase between them drifts over time at a rate comparable to the breathing signal. Bench tests confirmed that this drift, over a one-minute interval, exceeds the expected respiratory phase changes, making the setup impractical for vital-sign sensing without external clock synchronization.

B. BladeRF 2.0 Micro (rejected)

The BladeRF [9] has a 12-bit ADC and supports full-duplex operation on a shared LO, making it a strong candidate for this study. However, in the project's development environment-Windows Subsystem for Linux 2 (WSL2)-the USB 3.0 pass-through layer was unable to reliably sustain a 1 MSPS complex data stream. Buffer underflows occurred at approximately one-second intervals, leading to the loss of contiguous sample blocks and disruption of phase continuity in GNU Radio. Although migrating to a native Linux installation would likely resolve this limitation, it was not feasible within the project timeline. Consequently, the BladeRF was is retained as a candidate platform for future work pending a native Linux deployment.

C. ADALM-Pluto SDR (Final)

The ADALM-Pluto, based on the Analog Devices AD9363 integrated transceiver [10], [11], was selected as the delivered platform. It provides a 12-bit ADC, full-duplex operation on a single shared LO, guaranteeing phase coherence by construction- and streams samples to the host over an IP-over-USB or IP-over-Ethernet link, avoiding the USB-layer instability seen with the BladeRF. The nominal tuning range of 325 MHz to 3.8 GHz can be extended to approximately 6 GHz by firmware reconfiguration [12], which places the 5.95 GHz operating point within the usable band. Published LO phase noise is in the region of -90 dBc/Hz at 1 kHz offset, which is sufficient for respiratory detection but is the limiting factor for heart-rate recovery.

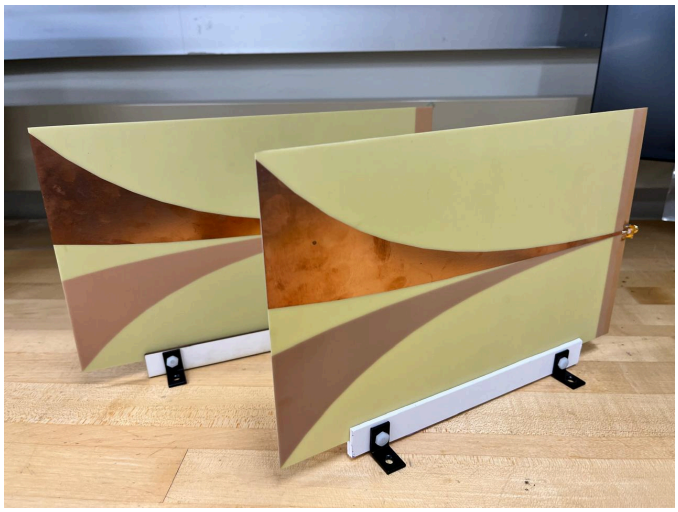


Fig. 2. Vivaldi antenna used for both TX and RX

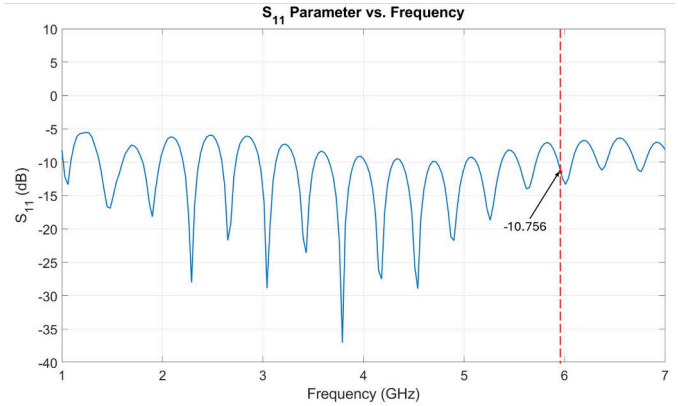


Fig. 3. Measured S11 of Vivaldi antenna used for both TX and RX. The marker at 5.95 GHz shows a return loss of 10.756 dB, sufficient for ISM band CW operation

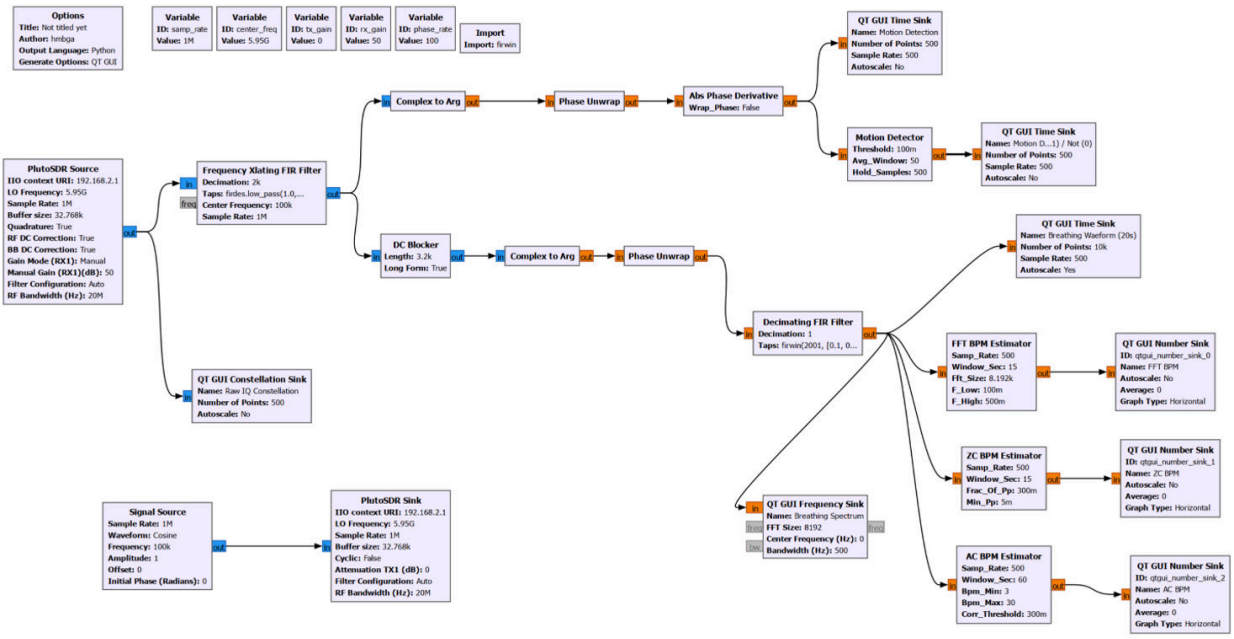


Fig. 4. Signal flow graph created by GNU Radio.

IV. Antenna Setup

The antenna setup was refined after initial experiments revealed issues with the received signal quality. A pair of log-periodic antennas was first used for transmission and reception; however, the resulting raw IQ constellation appeared distorted. This was traced to an impedance mismatch between the antenna feed and the SMA-to-coaxial adapter, which degraded signal integrity. To resolve this, the system was updated to use a properly matched pair of wideband Vivaldi antennas with standard SMA connectors which is shown in Fig 2. The operating frequency range of the Vivaldi antenna is from 1 GHz to 7 GHz. The matching performance was verified through S-parameter measurements as shown in Fig 3, confirming a return loss better than -10 dB across the operating frequency (with a measured value of -10.74 dB). Following this change, the received IQ constellation became clean and well-defined, enabling reliable downstream signal processing.

V. DSP Pipeline over GNU Radio

The digital signal processing (DSP) chain, implemented in GNU Radio 3.10, transforms the raw complex baseband data into a clean respiratory waveform. The raw 1 MSPS signal is first decimated by a factor of 2000 to a 500 SPS sample rate to focus on the low-frequency respiratory band. A 3200-tap complex DC blocker is then applied to the I/Q stream to eliminate static leakage from the transmitter, which would otherwise prevent accurate phase extraction. Following this, an $tan^{-1}(Q/I)$ operation and a custom streaming phase unwrap block recover the continuous phase signal. Finally, a 2001-tap Hamming window FIR bandpass filter (0.1–0.5 Hz) isolates the breathing signal, effectively suppressing high-frequency noise and lower-frequency drift.To maximize the reliability of the reported breaths-per-minute (BPM), the system employs a Consensus Method that aggregates data from three parallel estimators: Fast Fourier Transform (FFT), Zero-Crossing (ZC), and Autocorrelation (AC).

The FFT identifies the dominant spectral peak within a 20-second sliding window, while the ZC estimator provides low-latency tracking by counting cycles in the time domain. The AC estimator provides high steady-state accuracy by identifying the fundamental period of the signal via the Wiener-Khinchin theorem. A specialized voting block compares these three outputs; if any two estimators agree within a ± 3 BrPM tolerance, their mean is published as the official consensus (See fig. 4). This multi-modal approach allows the system to reject outliers caused by transient artifacts or filter ringing, ensuring that the final output is physiologically grounded and stable.

VI. Breaths Per Minute (BrPM) Estimation

Three estimators for respiratory rate operate concurrently on the bandpassed phase signal. Each has a distinct failure mode where the consensus layer fuses their outputs to reject individual-estimator errors.

A. Estimator Algorithms

1) FFT Peak estimator: A 20 s sliding window of the bandpass-filtered phase signal (10,000 samples at 500 SPS) is zero-padded to 8192 points and windowed using a Blackman–Harris function. The magnitude spectrum is then computed, and the dominant spectral component within the 0.1–0.5 Hz band is identified. To improve frequency resolution, parabolic interpolation is applied to the peak and its adjacent bins, reducing quantization error to below one bin and yielding an estimated rate uncertainty of approximately ±0.2 BPM.

2) Zero-crossing estimator: The bandpass-filtered phase signal is processed using a 15 s rolling window, after removing its mean, and zero crossings are detected. To reduce false crossings due to noise, a Schmitt-trigger-style hysteresis is enforced: only crossings associated with a local peak-to-peak amplitude greater than 30% of the window's overall peak-to-peak amplitude are retained. The respiratory rate is computed as $(N_{\text{crossings}} - 1) \times 60/(2\, t_{\text{window}})$ BPM, with the factor of two reflecting the two crossings per breathing cycle.

3) Autocorrelation estimator: A 60 s window of the bandpass-filtered phase signal is processed using autocorrelation computed by taking the inverse FFT of the magnitude-squared spectrum. The dominant peak within the lag range corresponding to 3–30 BPM is then identified, and the respiratory rate is estimated as $60/\tau_{\text{peak}}$ BPM, where τ_{peak} denotes the lag in seconds. To ensure reliability, a correlation coefficient threshold of 0.3 is applied; if no peak within the search range exceeds this threshold, the estimator declares no valid detection.

B. Consensus Voting

The three estimators publish BPM values at the 20 Hz dashboard refresh rate. A consensus block computes the three pairwise absolute differences between them and selects the smallest pair. If that pair's separation is within a tolerance of 3 BPM, their mean is published as the consensus rate and the agreement count is reported (3/3 if all three are within tolerance of the mean, 2/3 otherwise). If no pair is within tolerance, the output is suppressed and a diverging indicator is shown. This rule set eliminates the need to diagnose individual-estimator failure modes at runtime — the system simply reports readings only when at least two independent estimators corroborate each other (See fig. 6).

VII. Motion Detection

The motion-detection branch processes the phase-derivative signal $|\mathrm{d}\phi/\mathrm{d}n|$ using a three-stage gating scheme. First, a 50-sample (100 ms) moving-average filter implemented with a ring buffer suppresses high-frequency noise. Second, the smoothed signal is compared against a threshold of 0.1 rad/sample, corresponding to an approximate instantaneous chest velocity of 200 mm/s at 500 SPS. Third, a 500-sample (1 s) hysteresis hold is applied to prevent rapid toggling during brief pauses within a motion event. The resulting binary still/motion flag is used to gate the BPM display and to trigger event-based logging entries (See fig. 7).

VIII. Real-time GUI via PyQt5

The operator interface is a custom PyQt5 application built on pyqtgraph in Python, refreshed at 20 Hz (50 ms timer period). GNU Radio flowgraph transfer samples from the signal-processing threads to the GUI thread without inter-process communication overhead. The display panels include a raw IQ constellation (500-sample slice of the 1 MSPS stream), the bandpassed respiratory waveform over a 20-second window, three BPM gauges (one per estimator) and a consensus readout

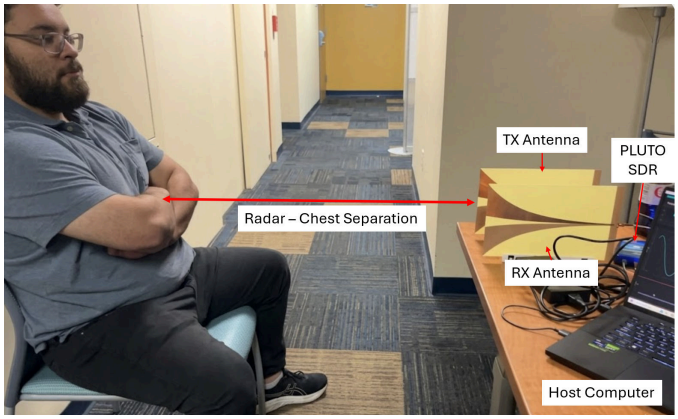


Fig. 5. Experimental setup test configuration



Fig. 6. VitaWave Dashboard when stable breathing pattern observed

with confidence indicator, a binary motion LED, and a time-series plot of the motion detector's continuous and gated outputs. Additionally, a session logger module records all events to a timestamped CSV file when the operator clicks a record button.

IX. Experimental Setup and Results

The overall experimental configuration is illustrated in Fig. 5. The transmit (Tx) and receive (Rx) radar antennas are positioned facing the chest of the subject to capture respiration-induced motion. After allowing a brief settling period for the system to stabilize, the processed outputs are visualized on the dashboard shown in Fig. 6, where the breathing waveform can be clearly observed. In addition to the respiration signal, the system also monitors subject motion, providing a real-time motion-detection indicator along with the instantaneous rate of change of phase. These outputs enable simultaneous observation of breathing activity and motion artifacts, facilitating reliable interpretation of the measured data.

X. Conclusions

This paper presents a complete and operational contactless respiratory-rate sensing system built using commodity SDR

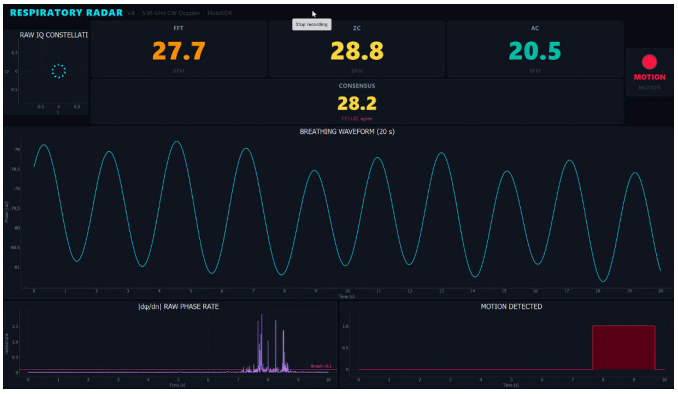


Fig. 7. Vitawave Dashboard when motion detected

hardware, along with a detailed characterization of its performance limits. The implemented system reliably measures respiratory rates in the 6–30 BPM range at a standoff distance of 0.5–2 m for a single stationary subject.

Robustness is achieved through consensus voting across three independent estimators, providing a calibrated confidence metric, while a Schmitt-trigger-based motion gate suppresses corrupted measurements. All signal processing is performed in real time on a commodity host, and a CSV-based session logger records BPM estimates and motion events for subsequent analysis.

XI. Future Directions

Several avenues exist for extending this work. The most immediate is heart rate extraction. The current system is limited by the ADALM-Pluto's LO phase noise (Õ0 dBc/Hz at 1 kHz offset), which is insufficient to resolve the smaller cardiac displacements (0.2–0.5 mm vs. 5–12 mm for respiration). Migrating to a lower-noise transceiver such as the BladeRF 2.0 Micro on a native Linux host, or adding an external TCXO reference, could bring cardiac motion above the noise floor. A second direction is multi-subject sensing. The current architecture assumes a single stationary subject in the beam. Extending to spatially separated subjects would require either a phased-array front end for beam steering or a frequency-modulated continuous wave (FMCW) architecture that provides range resolution, allowing simultaneous monitoring of multiple patients at distinct distances — a meaningful step toward ICU or sleep-lab deployment.Third, embedded and edge deployment deserves attention. The entire DSP chain currently runs on a commodity host computer. Porting the GNU Radio flowgraph to a Raspberry Pi 5 or Jetson Nano, or re-implementing the consensus estimators as a bare-metal firmware on a microcontroller paired with the AD9363, would produce a self-contained, low-power device suitable for home or field use.Fourth, the current motion gate uses a simple threshold on the phase derivative. A more sophisticated approach — such as a hidden Markov model or a lightweight neural classifier trained on labeled still/motion segments — could distinguish between bulk body movement and deliberate deep breathing, reducing false suppression events during restless sleep monitoring.Finally, clinical validation against a certified reference (e.g., a capnograph or a piezoelectric respiration belt) across a cohort of subjects in varying postures (supine, seated, lateral) and at the outer boundary of the stated 2 m range would provide the quantitative accuracy and repeatability data needed to assess the system's suitability for real clinical scenarios.

XII. Acknowledgement

The authors thank the faculty of the EEE 4550/5557 Radar Systems course and Dr.Satheesh Bojja Venkatakrishnan at Florida International University for the initiative and providing the SDR hardware and antennas used in this work.

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